

Md Shahid Ali Khan

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PROFILE SUMMARY

Software Engineer with **3+ years of experience** in **Java and Spring Boot**, developing **microservices and distributed systems** skilled in **Kafka, Redis, concurrency, asynchronous processing, and backend performance optimization**.

TECHNICAL SKILLS

Languages & Core CS: Java 17, SQL, C++, Data Structures & Algorithms (DSA), OOP

Backend Technologies: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices

Messaging & Caching: Apache Kafka, Redis, Caffeine, Asynchronous Processing

Databases & Analytics: PostgreSQL, MySQL, ActivePivot

Cloud & DevOps: AWS (Lambda, S3, EC2, CloudWatch), Docker, GitHub Actions, CI/CD Pipelines, Maven

Architecture & Practices: Distributed Systems, System Design, Design Patterns, Concurrency, Multithreading, Resilience4j, Performance Optimization, Agile/Scrum

Testing & Tools: JUnit, Mockito, Git, IntelliJ IDEA, Postman

WORK EXPERIENCE

Société Générale

Software Engineer

Bangalore, India

July 2023 – Present

Financial Risk Reporting Platform | *Java, Spring Boot, Microservices, Apache Kafka, PostgreSQL, Spring Security, Caffeine*

- **Developed and enhanced Java 17/Spring Boot microservices and REST APIs** for a financial risk and liquidity reporting platform supporting NPV, NIM, GAP, sensitivity analysis, dashboards, exports, and regulatory reporting.
- **Developed analytical reporting workflows using RAFAL Java-based queries and ActivePivot OLAP**, defining reusable dimensions, measures, and filters; integrated with the illiqa-bigdata relay service through HTTP/JSON and implemented streaming response processing for **50K+ record** report workloads.
- **Implemented and optimized Caffeine L1 caching** through cache-key optimization, TTL configuration, and eviction policies, reducing frequently accessed API response times from approximately **10 s to 20 ms** and report generation time from **25 s to 10 s**.
- **Designed extensible report execution flows using Strategy and Factory patterns**, encapsulating report-specific orchestration and centralizing runtime flow creation. Used Builder-based construction for complex reporting DTOs and query models.
- **Implemented Kafka-driven asynchronous processing** for long-running report generation using producer-consumer workflows, concurrent workers, scheduled jobs, and dedicated thread pools to prevent API blocking and improve responsiveness.
- **Improved workflow reliability and fault isolation** through Kafka retry handling, idempotent processing, automated failure email notifications, and Resilience4j Circuit Breaker for downstream failure isolation.
- **Contributed to secure production delivery** using Spring Security, OAuth2/OIDC, JWT-bearer delegation, Spring Data JPA, Hibernate, and GitHub Actions CI/CD.
- Resolved production defects, participated in code reviews and knowledge transfer, contributed to Angular enhancements, and mentored **2 interns who converted to full-time employees**.

Software Developer Intern

March 2023 – July 2023

Rejected Transaction Impact Analysis & Script Migration

- Built a **Spring Boot**-based analysis platform to categorize rejected transactions by risk, processing **100K+ transaction records** using optimized SQL queries to accelerate remediation workflows.
- Migrated **70+ legacy Unix shell scripts** to Linux and refactored workflows, improving system compatibility, maintainability.

PROJECTS

Distributed Workflow & Job Processing Platform | *Java, Spring Boot, PostgreSQL, Kafka, Redis, Resilience4j, Docker*

- Built an **asynchronous job-processing platform** to execute **long-running tasks without blocking API requests**, using persistent **job lifecycle management**, Kafka consumer groups, **atomic state transitions**, and **concurrent workers**.
- Implemented the **Transactional Outbox pattern** to reliably propagate database changes to Kafka, preventing **dual-write inconsistencies** when event publishing fails.
- Implemented **Caffeine + Redis L1/L2 caching** using the **Cache-Aside pattern** to reduce repeated data access, and built **Redis-based distributed rate limiting** using **atomic Lua operations** for consistent request-throttling decisions across instances.
- Engineered **at-least-once Kafka processing** with acknowledgements, offset management, **atomic job claiming**, and **idempotency** to handle duplicate message delivery safely.
- Implemented resilient job execution with **exponential backoff**, **nextAttemptAt-based retries**, Kafka error handling, and **Dead Letter Topics (DLT)** to support recovery from transient failures and isolate unrecoverable messaging failures.
- Implemented **Redis distributed locking with expiry** to coordinate job execution across application instances, and applied **Resilience4j Circuit Breaker** patterns to handle Redis dependency failures gracefully.

EDUCATION

Birla Institute of Technology, Mesra (BIT Mesra)

Ranchi, India

B.Tech in Information Technology | GPA: 8.60/10

2019 – 2023

- Solved **650+ Data Structures & Algorithms problems**, demonstrating strong problem-solving and algorithmic skills.
- Secured **National Rank 844** in CODEKAZE 2021 among **200,000+ participants**.
- Received **“Spot” and “Newbie” Awards** at Societe Generale for strong performance and contributions.